

Individual differences in skilled adult readers reveal dissociable patterns of neural activity associated with component processes of reading

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ABSTRACT

We used fMRI to examine patterns of brain activity associated with component processes of visual word recognition and their relationships to individual differences in reading skill. We manipulated both the judgments adults made on written stimuli and the characteristics of the stimuli. Phonological processing led to activation in left inferior frontal and temporal regions whereas semantic processing was associated with bilateral middle frontal activation. Individual differences in reading subskills were reflected in differences in the degree to which cortical regions were engaged during reading. Variation in sight word reading efficiency was associated with degree of activation in visual cortex. Increased phonological decoding skill was associated with greater activation in left temporo-parietal cortex. Greater reading comprehension ability was associated with decreased activation in anterior cingulate and temporal regions. Notably, associations between reading ability and neural activation indicate that brain/behavior relationships among skilled readers differ from patterns associated with dyslexia and reading development.

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1. Introduction

Successful reading in adults involves translating a visual code into both sound-based (phonological) and meaning-based (semantic) codes; as a result, it requires coordinating a range of mental processes. Consistent with this, advances in neuroimaging have uncovered a reading network spanning wide-ranging cortical regions. A prevailing view holds that different subcomponents of this network correspond to processes involved in accessing orthography (i.e., visual symbols and the ways in which they can be combined), phonology (i.e., sounds of language), and semantics (i.e., word meanings; (Devlin, Matthews, & Rushworth, 2003; Fiez, 1997; Price, Moore, Humphreys, & Wise, 1997; Pugh et al., 1996; Rumsey et al., 1997). In parallel with this, a separate line of research has sought to link patterns of brain activity to individual differences in reading subskills. However, the focus of these studies has been the trajectory of neural changes reflecting normal development of reading (Church, Coalson, Lugar, Petersen, & Schlaggar, 2008; Turkeltaub, Gareau, Flowers, Zeffiro, & Eden, 2003) or wide-scale neural differences in individuals with dyslexia (Brunswick, McCrory, Price, Frith, & Frith, 1999; Temple et al., 2001). In the current study we aimed to bridge these lines of research by examining

how individual differences in specific reading skills among a wide range of adults relates to activation in specific brain regions engaged by reading.

1.1. Component processes of reading

Studies of reading and the brain have emphasized a group of cortical regions that are engaged during reading. Moreover, stimulus properties and task demands appear to differentially modulate activity in these regions, such that different hypotheses have been put forward concerning their function. Here we summarize some of the key findings in this regard.

The first of these is the proposal that the left ventral temporal region, including ventral extrastriate cortex and the fusiform gyrus, supports pre-lexical orthographic analysis during reading (Cohen et al., 2000). This type of analysis includes not only the recognition of individual orthographic units, but also the knowledge of how these units can be legally combined to form valid words in a language. This allows for relatively efficient recognition of familiar words, despite variations in physical characteristics such as font and case (McCandliss, Cohen, & Dehaene, 2003). Evidence that the midfusiform gyrus is involved in this type of processing comes from observations that this region shows greater activation to orthographically legal letter strings than to either nonalphabetic stimuli (Price, Wise, & Frackowiak, 1996; Puce, Allison, Asgari, Gore, & McCarthy, 1996) or to strings of letters that are orthographically illegal (Buchel, Price, & Friston, 1998; Cohen et al.,

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2002). Additionally, activity in this region is modulated during tasks that place specific demands on orthographic processing such as case judgment (Pugh et al., 1996), cross-case repetition priming (Dehaene et al., 2004), lexical decision with anagrams (Pammer et al., 2004) and reading Chinese logographic characters versus Chinese words presented in an alphabetic script ('pinyin'; (Chen, Fu, Iversen, Smith, & Matthews, 2002). Left fusiform activation in response to written language is greater in literate than illiterate adults (Dehaene et al., 2010), even when literacy was acquired in adulthood, suggesting that this region is sensitive to experience. The fusiform gyrus is also active during written and auditory spelling, which require access to word form information, than during auditory and written rhyme tasks, which do not (Booth et al., 2002). Involvement of this area in visual word form analysis is consistent with the broader role of the inferior temporal cortex in object recognition and visual expertise (Dehaene & Cohen, 2011; McCandliss et al., 2003), although it remains hotly debated whether there exists a cortical specialization for visual words to the exclusion of other orthographic stimuli and other non-visual language processing (Price & Devlin, 2003; Price & Devlin, 2011). For example, priming with picture/word pairs that do not have similar visual forms reduces activation in the fusiform gyrus (Kherif, Josse, & Price, 2011), which has been taken as evidence that this region is sensitive to top-down influences.

A second set of findings highlights the roles of left inferior frontal cortex and left temporo-parietal regions in phonological processing. Pugh and colleagues have found greater activity for rhyme judgments versus letter case judgments in the left inferior frontal gyrus (IFG) and left superior temporal gyrus (STG) (Pugh et al., 1996). Left IFG has been found to be differentially active during phonological analysis when compared with a picture-matching task using the same stimuli (Katzir, Misra, & Poldrack, 2005). Additionally, activation in the left frontal operculum is modulated by word frequency and spelling-to-sound consistency (reviewed in Fiez and Petersen (1998)). Attending to sound similarity while reading lists of phonologically related words led to greater activation in posterior portions of the left IFG than attending to meaning relationships while reading lists of semantically related words (McDermott, Petersen, Watson, & Ojemann, 2003). Similarly, left temporo-parietal activation has been associated with syllable counting (Price et al., 1996) and data from Chinese reading indicate that reading pinyin yields greater activation in bilateral inferior parietal and superior temporal cortex than reading Chinese logographic characters, due to the greater involvement of phonological decoding in reading pinyin (Chen et al., 2002). The exact roles that left IFG and temporo-parietal regions play in phonological processing remains somewhat unclear. It has been suggested that IFG activity reflects demands placed on working memory by phonological segmentation or articulatory recoding processes (Burton, Locasto, Krebs-Noble, & Gullapalli, 2005), with activation of temporo-parietal cortex reflecting grapheme-to-phoneme conversion (Blomert & Froyen, 2010; Jobard, Crivello, & Tzourio-Mazoyer, 2003). Finally, inferior frontal activation may reflect implicit access of articulatory representations (Hickok & Poeppel, 2004; Zatorre, Meyer, Gjedde, & Evans, 1996). While this model is typically applied to studies of auditory speech processing, one might also assume that this group of cortical regions represents phonological processing that abstracts away from either the visual or auditory modality.

A more complex set of regions appears to be engaged in processing words' meanings. Studies of patients with semantic dementia suggest a role for anterior temporal regions in semantic processing (Mummery et al., 2000), though the exact role this region plays in this regard is still under debate (Simmons & Martin, 2009). The semantic network likely also includes portions of the left middle temporal cortex, as this region appears sensitive to

semantic judgment compared to phonological or orthographic judgments (Price et al., 1997; Pugh et al., 1996; Vandenberghe, Price, Wise, Josephs, & Frackowiak, 1996). Words with greater familiarity, meaningfulness, and number of semantic features more strongly engage the angular gyrus and posterior cingulate bilaterally, perhaps reflecting increased semantic processing of these items (Graves, Desai, Humphries, Seidenberg, & Binder, 2010). Notably, studies have also identified increased activity in portions of left inferior frontal cortex while processing semantic characteristics of words (Bookheimer, 2002; Poldrack et al., 1999). This includes living/nonliving judgments (Kapur et al., 1994), abstract/concrete decisions (Demb et al., 1995; Gabrieli et al., 1996), and verb generation (Petersen, Fox, Posner, Mintun, & Raichle, 1988). Importantly, the portion of IFG involved in semantic processing appears to lie in a distinct subregion that is anterior to the one engaged during phonological processing (Gough, Nobre, & Devlin, 2005; McDermott et al., 2003).

In spite of a considerable literature concerning the neural mechanisms related to different components of word knowledge, prior studies do present some limitations. Specifically, some studies have focused exclusively on individual component processes of visual word recognition (e.g., Cohen et al., 2000) or on how specific cortical regions respond to different word types or tasks (e.g., Graves et al., 2010). While these lines of research have been highly instructive, it can be difficult to extrapolate across this type of study to obtain a full picture of the patterns of neural activity during reading. Other studies have taken on the goal of differentiating different component processes of reading via task decomposition (Chen et al., 2002; McDermott et al., 2003; Pugh et al., 1996; Rumsey et al., 1997), taking the approach of isolating subprocesses of visual word recognition by varying both stimuli and tasks. However, this could have the effect of confounding differences in processing with differences in stimulus characteristics.

In order to investigate component processes of visual word recognition without confounding task differences and stimulus differences, the present study takes a different approach of asking subjects to perform different tasks while maintaining stimulus properties across conditions. This approach is similar to prior studies focusing on phonological versus semantic processing of familiar English words (Devlin et al., 2003; Price & Devlin, 2003; Price et al., 1997), but with additional conditions that allowed us to isolate brain regions related to orthographic processing as well.

1.2. Individual differences in reading

A separate line of research has investigated relationships between individual differences in brain activation and individual differences in reading skill. This work has typically included comparisons of individuals with reading impairment to typical readers, or tracked changes in activation associated with reading development, with a smaller number of studies focusing instead on individual differences within typical adult readers. Below we discuss the evidence for associations between such individual differences and activation associated with specific reading skills.

We first consider individual differences in the ventral portion of left inferior temporal gyrus, a region associated with orthographic processing. There is conflicting evidence regarding relationships between reading skill and activation in this region. In a magnetic source imaging study, children with dyslexia showed similar activation profiles to age-matched non-impaired readers in the left basal temporal areas (approximately BA 37) during printed word and pseudoword reading (Simos, Breier, Fletcher, Bergman, & Papanicolaou, 2000a). However, adolescents with dyslexia showed reduced activation in the fusiform gyrus bilaterally during visual rhyme judgments when compared to both reading and age matched groups (Hoeft et al., 2007). Among children aged 13–15, increased

activity in the fusiform gyrus was associated with smaller gains in reading fluency (McNorgan, Alvarez, Bhullar, Gayda, & Booth, 2011). College students with dyslexia showed less activation than similar-aged controls in this region during reading aloud and implicit reading tasks (Brunswick et al., 1999). Similarly, positive correlations of decoding ability with right fusiform activation during rhyme compared to rest were observed among children identified as struggling readers by their teachers (Hoeft et al., 2007). In contrast, in a PET study, typical adult readers' letter recognition accuracy was negatively related to metabolism in the left inferior temporo-occipital cortex (Garrett et al., 2000). Improvements in performance on a measure of Sight Word Efficiency were associated with increased cortical sensitivity to word visibility near the visual word form area (Ben-Shachar, Dougherty, Deutsch, & Wandell, 2011). Among adults, but not children, stronger reading skills were associated with greater coupling of fusiform and inferior frontal and inferior parietal regions (Koyama et al., 2011). Given that the left fusiform region has been proposed to support orthographic analysis (Cohen et al., 2000), it is possible that individual differences in degree of activation might relate differently to reading tasks that place differential demands on orthographic processing.

Similar controversy surrounds relationships between inferior frontal activation and reading skill. Children with dyslexia showed less activation in the left IFG during pseudoword reading and phoneme awareness than controls (Georgiewa et al., 1999). Similarly, in children and young adults, reading ability was positively correlated with activity in the left ventral inferior frontal gyrus (Turkeltaub et al., 2003) and increased activity in the inferior frontal gyrus was associated with increases in reading fluency among children aged 9–11 (McNorgan et al., 2011). In contrast, dyslexic college students showed greater activation of pre-motor Broca's area (BA 6/44) than controls while reading aloud. The left inferior frontal gyrus also showed greater activation during rhyme judgment in adults with dyslexia (MacSweeney, Brammer, Waters, & Goswami, 2009). These conflicting results may reflect the multiple functions thought to be subserved by the IFG, and its potential involvement in compensatory processing.

The picture is somewhat clearer for posterior temporo-parietal cortex in which studies have typically found that greater activation is associated with better performance of phonological tasks. Reduced activation of the left temporo-parietal areas during phonological processing has been observed in children with (or at risk for) dyslexia (Papanicolaou et al., 2003; Simos et al., 2000b; Temple et al., 2001) and in compensated dyslexic adults (Paulesu et al., 1996). Phonological awareness was positively correlated with left perisylvian activity during a one-back task with pronounceable nonwords (Noble, Wolmetz, Ochs, Farah, & McCandliss, 2006). Greater phonological decoding ability was associated with rhyme-related activation in the left middle temporal gyrus among children (Hoeft et al., 2007). Dyslexic adolescents showed less activation in left parietal cortex than both age and reading ability-matched groups (Hoeft et al., 2006; Hoeft et al., 2007), which has been taken as evidence that the reduction is characteristic of dyslexia rather than reading skill. However, in typical adult readers, better auditory and visual rhyme performance was associated with greater activation in superior temporal and supramarginal/angular gyri (Booth et al., 2002), suggesting that these relationships may extend beyond impaired reading. In a sample including both dyslexic and skilled readers, greater activation in bilateral parietotemporal regions during semantic judgments was associated with better performance on a standardized measure of pseudoword reading (Shaywitz et al., 2002).

Finally, there is some indication that individual differences in activation in other regions, which appear to be associated with semantic processing, may also be correlated with reading skill.

Among young adults, greater reading comprehension was associated with greater activation during the processing of written anomalous sentences in inferior and middle frontal gyri (Shankweiler et al., 2008). Among children, word and nonword reading ability were positively associated with activation in the left posterior MTG, right inferior parietal lobule, and left postcentral gyrus (Meyler et al., 2007). The development of reading ability over time correlated positively with activity in the left middle temporal gyrus (MTG) (Turkeltaub et al., 2003). Children who received reading intervention also showed increased activity and reduced latency of onset of activity in the posterior superior temporal region while reading aloud sight words (Simos et al., 2007). However, such results are complicated by the possibility that changes over the course of reading development or reading intervention relates to developing phonological, rather than semantic processing abilities. Similarly, it has been hypothesized that observed developmental decreases in temporo-parietal activation during reading of high-frequency words and repetition of auditorially presented words relates to differential reliance on phonological processing (Schlaggar & Church, 2009). Given the importance of general semantic and conceptual knowledge in understanding connected texts, and the relative preservation of these skills in dyslexia, studies of reading impairment may not be the most effective way to investigate neural correlates of semantic processing skill.

A key motivation for the present study relates to the observation that prior studies of individual differences have typically examined neural correlates of overall reading ability rather than attempting to isolate different behavioral markers of reading skill (e.g., phonological decoding, orthographic awareness, vocabulary and conceptual knowledge, comprehension of connected text, to name a few). As such, it is not clear precisely how observed variation in patterns of brain activation differentially relate to orthographic, phonological, or semantic skill. Additionally, such studies are marked by conflicting results, likely due to differences in subject populations and varying methodologies in isolating neural activity related to reading subskills. It likewise remains an open question whether atypical patterns of activation shown in dyslexia are mirrored by individual differences within a normal reading population, or instead reflect qualitative differences in the brain mechanisms supporting reading in impaired individuals. Finally, developmental studies may conflate reading skill and experience with brain maturation, making it difficult to know whether differences between younger and older readers are congruent with individual differences among adult readers.

1.3. Overview of the present study

In the current study, we sought to tease apart the neural underpinnings of visual word recognition by manipulating both the judgments participants made on pairs of stimuli and the characteristics of these stimuli. Adult readers across a wide range of ages and reading ability performed case judgments, rhyme judgments and category judgments on pairs of stimuli. For case judgments, stimuli included real English words, pronounceable pseudowords, and consonant strings, allowing us to contrast brain activation elicited by familiar words to that elicited by novel strings to investigate neural substrates of orthographic processing. Rhyme judgments made on pairs of pseudowords and words were contrasted with case judgments on the same stimuli to investigate patterns of activation associated with phonological processing. We contrasted activity during meaning judgments to case and rhyme judgments performed on identical stimuli to investigate activity associated with semantic processing. For the orthographic contrast, we predicted that activation would be greater for English words than consonant strings in the left fusiform gyrus on the basis that this area has been proposed to support orthographic analysis

(Cohen et al., 2000). Left posterior inferior frontal and superior temporal areas were predicted to be more active during rhyme judgments than case judgments as these areas show increased activity during phonological analysis (Pugh et al., 1996). Finally, we anticipated that left temporal and frontal areas would show greater activation during meaning judgments than orthographic and phonological tasks. Because identical stimuli were presented for the latter contrasts, differences in activation pattern are likely to result from the different cognitive demands of the task rather than differences in the properties of stimuli.

To investigate relationships between individual differences in reading skills and patterns of neural activity, brain activation associated with reading familiar words was correlated with standardized reading measures performed outside of the scanner. We explored relationships between different reading skills and activation within the cortical regions engaged by reading by correlating performance on different standardized reading measures with brain activation for word case, rhyme, and meaning judgments. Thus, we aimed to identify associations between activation in different component brain regions and skill at different reading tasks. Importantly, we used statistical maps encompassing all word-related tasks rather than the three finer-grained contrasts in order to avoid biases that might have been introduced by correlating individual behavioral measures with their corresponding subnetworks. This approach maximized our ability to identify individual difference correlations across the entire set of regions engaged by reading, on the logic that differences in reading subskills might have wide-reaching impacts that extend across multiple areas of the brain. If the same cortical regions show different patterns of correlation with different reading subskills, this suggests that different cortical regions may play different roles in skilled reading.

2. Methods

2.1. Participants

Twenty neurologically normal adults (13 female) took part in the study. All were right-handed native speakers of English recruited from newspaper advertisements in the London, Ontario community and postings at the University of Western Ontario. Demographic data are presented in Table 1. Scores on the reading measures ranged from below average to well above average, representing the variability present in a community sample.

2.2. Psychometric measures

Prior to the scanning session, all participants completed a battery of intelligence and reading measures. Nonverbal intelligence was assessed with the Test of Nonverbal Intelligence-3, which uses picture stimuli to evaluate skill at recognizing logical sequences (TONI-3; (Brown, Sherbenou, & Johnsen, 1997). The Sight Word Efficiency (SWE) and Phonemic Decoding Efficiency (PDE) subtests of the Test of Word Read Efficiency (TOWRE) were used to assess timed reading of familiar words and pronounceable pseudowords,

respectively (Torgesen, Wagner, & Rashotte, 1999). Form G of the Nelson–Denny Reading Comprehension subtest (Brown, Fishco, & Hanna, 1993), in which comprehension questions follow short passages, was used as a timed measure of reading comprehension. There was a significant association between performance on the SWE subtest and reading comprehension ($R = .44$, $p < .05$). Performance on the PDE subtest was not significantly correlated with either of the other two reading measures (PDE with SWE, $R = .37$; PDE with Nelson–Denny, $R = .08$).

2.3. fMRI tasks

Participants performed orthographic, phonological, and semantic judgments on pairs of letter strings. The visual stimuli were presented one above the other on a screen located at the foot of the scanner bore. A vertical arrangement was used to minimize unintended effects of visual field bias on resulting behavioral or fMRI data; this procedure also minimized eye movements correlated with stimulus presentation, which might have introduced artifacts in fMRI results. For *case decisions*, participants judged whether the pair of stimuli showed the same pattern of upper and lower case letters (e.g., *cOmB-rOAm*). This task was performed on consonant strings, pronounceable pseudowords or English words. For *phonological decisions*, participants decided if pairs of stimuli (pronounceable pseudowords or English words) rhymed. For half of the rhyming pairs, the spelling was dissimilar to discourage decisions based on orthography (e.g., *comb-roam*). For *semantic decisions*, participants viewed only familiar English words, and judged whether both were types of animals. Each pair was presented for 2.5 s, and participants were encouraged to make a match/no match decision as quickly and accurately as possible. In each of the six conditions, half of the pairs matched. Participants responded with a match/no match button press on each trial. Behavioral results are presented in Table 2.

Identical word stimuli, in the same pattern of upper and lower case, were used across orthographic, phonological, and semantic tasks, and the same pseudoword stimuli were used for orthographic and phonological decisions. Thus, participants saw the same stimuli when making case, rhyme, and meaning judgments. This repetition, in which a word pair was viewed three times during the hour of scanning, was necessary in order to maintain tight control over the psycholinguistic properties of individual words. However, task order was counterbalanced so that the influence of repetitions was equal across each task type.

Each task was presented in blocks of seven trials with order of trials randomized within blocks. Participants were cued as to which task to perform in an upcoming block by an instruction screen presented for 2.5 s before the start of the block. Each trial was presented until participants responded, then replaced by a fixation cue (+) at the center of the screen, for a total length of 2.5 s per trial (trials on which no response was provided were coded as errors). Each 17.5 s block was followed by a rest period of 10 s during which a fixation cue was presented. Scanning was divided into separate runs of 168 trials, which minimized fatigue by providing rest periods. There were a maximum of 8 runs, with all

Table 1
Demographic characteristics of the sample ($N = 20$).

	Mean (SD)	Range
Age	29.7 (13.7)	19–59
Years of education	15.1 (2.1)	12–18
TONI (scaled score)	96.4 (17.2)	72–140
TOWRE SWE (scaled score)	95.9 (7.4)	83–111
TOWRE PDE (scaled score)	94.6 (9.9)	81–120
Nelson–Denny Comprehension	219.2 (23.3)	163–251
Nelson–Denny reading rate	219.4 (32.7)	171–290

Table 2
Mean (standard deviation) performance on each task.

	Accuracy (percent correct)	RT (ms)
Consonant string case	86.5 (11.1)	1217 (182)
Pseudoword case	81.4 (22.3)	1208 (163)
Word case	83.1 (14.9)	1211 (160)
Pseudoword rhyme	86.7 (10.6)	1158 (169)
Word rhyme	93.6 (5.9)	1097 (216)
Word meaning	79.4 (11.9)	1180 (129)

subjects completing at least 6 runs, and 15 subjects completing all 8 runs. Two blocks of each trial type were presented per experimental run, with block order randomized within runs.

2.4. Stimuli

Familiar word stimuli ranged in length from 3 to 8 letters. Pairs of words were selected so that they matched along either the orthographic, phonological or semantic dimensions; care was taken to equate these matching pairs to non-matching pairs with respect to length (number of letters), log printed frequency (Kucera & Francis, 1967), number and frequency of orthographic neighbors (Medler, 2005), and bigram frequency (Medler & Binder, 2005). Items for the pseudoword trials were constructed by changing a single letter from each word. The resulting pseudowords remained pronounceable in English, and matched the words in number of letters, number and frequency of orthographic neighbors, and bigram frequency. Non-pronounceable consonant strings were constructed by replacing the vowels in the pseudowords with consonants. Stimulus characteristics are presented as [Supplementary material \(Table S1\)](#).

2.5. fMRI acquisition and processing

Imaging was performed on a 3 T Siemens Tim Trio scanner equipped with a 32-channel head coil for transmit/receive. Functional scans consisted of 1152 T2*-weighted EPI volumes acquired in an axial orientation (TR = 2.5 s, TE = 30 ms, flip angle = 90°, FOV = 240 mm in-plane, 80 × 80 × 36 matrix, voxel size = 3 × 3 × 3.5 mm) using an iPAT parallel acquisition sequence (generalized auto-calibrating partially parallel acquisition [GRAPPA]; acceleration factor = 2). A conventional T1-weighted anatomical scan was used to select the location of functional volumes, which were oriented in a way that covered the entire cerebrum (excluding the most inferior portions of the cerebellum). A 3D T1-weighted anatomical scan was acquired mid-session (MPRAGE; GRAPPA acceleration factor = 2; TR = 2.3 s, TE = 2.97 ms, Flip angle = 9°, 256 × 240 × 192 matrix; voxel size = 1 × 1 × 0.87 mm). Motion correction of functional scans was performed using an automatic correction procedure during scanning, registered to the first scan of each sequence. Following this, the AFNI software package (Cox, 1996) was used to register each volume to the volume acquired immediately following the anatomical scan, using a 3d rigid body transformation (AFNI *3dvolreg*). Correction for slice scan timing was also adjusted offline (AFNI *3dTshift*, quintic interpolation).

Subject-wise functional scans were transformed to percent signal change and analyzed using a general linear model (GLM; AFNI *3dDeconvolve*) with six predictors: consonant string case decisions, pseudoword case decisions, pseudoword rhyme decisions, word case decision, word rhyme decisions, and word meaning decisions. Each predictor was modeled as a boxcar function assuming a block duration of 17.5 s convolved with a canonical hemodynamic response function. Because performance was relatively high, both correct and incorrect trials were analyzed. Each subject's anatomical scan was coregistered to the standard 3D space of Talairach and Tournoux (1988) using an automatic rigid body transformation procedure; this transformation was subsequently applied to each subject's statistical maps, which were then resampled to 1 mm³ and spatially filtered using a 5 mm FWHM Gaussian kernel.

Groupwise analyses examined the following statistical contrasts: (1) to identify regions associated with processing low-level visual characteristics of orthographic stimuli we performed a conjunction analysis that identified regions activated by both the WORD CASE > WORD RHYME and PSEUDOWORD CASE > PSEUDOWORD RHYME contrasts; (2) to identify neural responses to orthographically legal vs. illegal strings we performed a conjunction

analysis that identified regions that are jointly activated by the WORD CASE > CONSONANT STRING CASE and PSEUDOWORD CASE > CONSONANT STRING CASE contrasts; this analysis was intended to identify regions engaged by processing of word-level orthographic information regardless of lexical status, while abstracting away from more basic visual processes that are related to single-letter identification; (3) to identify regions engaged by phonological processing, we performed a conjunction analysis that found regions commonly activated by the WORD RHYME > WORD CASE and PSEUDOWORD RHYME > PSEUDOWORD CASE contrasts; (4) to find regions associated with semantic processing, the conjunction analysis examined regions commonly activated by the WORD MEANING > WORD CASE and WORD MEANING > WORD RHYME contrasts. These contrasts examined relative activation in all voxels and were not restricted to voxels that differed significantly from rest. Thus, it is possible that we identified some regions that were jointly deactivated, relative to rest, albeit with one condition showing a lesser degree of this relative deactivation. Tables which distinguish between regions that show significantly greater activation than rest and regions that do not show greater activation than rest are presented as [Supplementary material \(Tables S2–S4\)](#). Groupwise statistical maps were created by entering subjectwise parameter estimates for each contrast of interest into one-sample *t*-tests. Correction for multiple comparisons was obtained using a Monte Carlo simulation (AFNI AlphaSim, 10,000 iterations), yielding a corrected alpha level of .05 at a minimum contiguous cluster size of 243 mm³ (based on an uncorrected threshold of $p < .001$).

We explored relationships between reading skill and functional activation by conducting correlational analyses. In order to reflect activation during orthographic, phonological, and semantic processes equally, only activation during word trials (for which semantic content is available) were included in this analysis. Subjects' mean beta values across the three word trial conditions (WORD CASE, WORD RHYME, WORD MEANING) were regressed against the three off-line behavioral measures of interest (AFNI *3dRegAna*). This provided a picture of the portions of the cortical regions engaged during reading that predicted specific reading sub-skills. The same correction factor was applied for multiple comparisons as above.

3. Results

3.1. Component processes of reading

The first analysis investigated activation associated with low-level visual characteristics of orthographic stimuli. We computed statistical maps for the WORD CASE > WORD RHYME and PSEUDOWORD CASE > PSEUDOWORD RHYME contrasts as well as the conjunction of the two (Fig. 1A). This yielded a large cluster in the right occipital region and smaller clusters in left occipital cortex and bilateral frontal cortex, which were significant for both contrasts (Table 3). We also sought out regions showing a specialization for processing orthographically legal stimuli versus illegal consonant strings, using a conjunction analysis for WORD CASE > CONSONANT STRING CASE and PSEUDOWORD CASE > CONSONANT STRING CASE. Here, no clusters were observed for the conjunction. We did, however, observe a significant cluster in the left middle temporal gyrus (MTG) for PSEUDOWORD CASE > CONSONANT STRING CASE; likewise, a significant cluster in the frontal region was significant for WORD CASE > CONSONANT STRING CASE (Fig. 1B).

The next analysis investigated activation associated with phonological processing, which we computed by the conjunction of WORD RHYME > WORD CASE and PSEUDOWORD RHYME > PSEUDOWORD CASE contrasts. This yielded a number of clusters

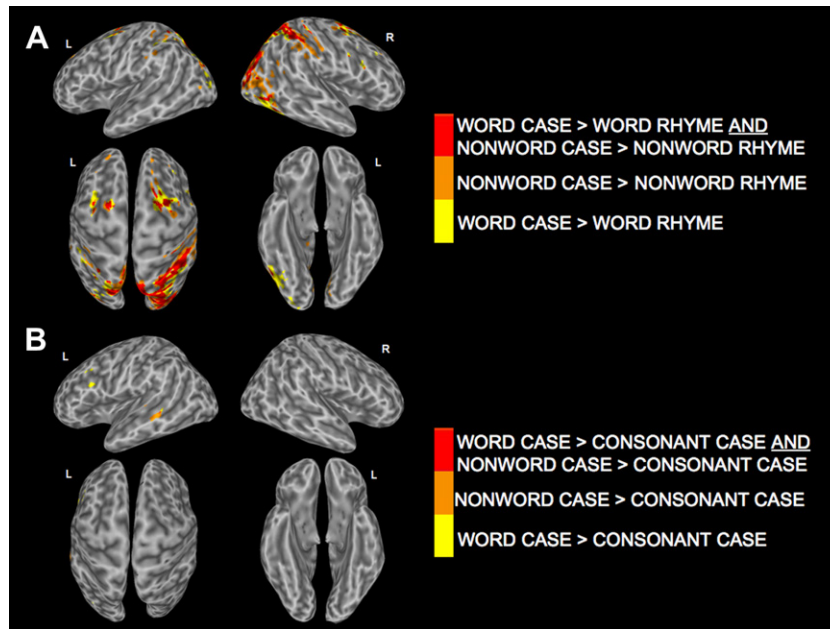


Fig. 1. (A) Regions showing greater activation in WORD CASE > WORD RHYME and PSEUDOWORD CASE > PSEUDOWORD RHYME contrasts ($p < .005$) displayed on a cortical surface model of the N27 atlas brain. (B) Regions showing greater activation in WORD CASE > CONSONANT STRING CASE and PSEUDOWORD CASE > CONSONANT STRING CASE contrasts ($p < .005$) displayed on a cortical surface model of the N27 atlas brain.

Table 3

Regions demonstrating greater activation during WORD CASE > WORD RHYME and PSEUDOWORD CASE > PSEUDOWORD RHYME contrasts.

Anatomical location		Talairach coordinates ^a			Size (mm ³)
		x	y	z	
Temporal	R Sup. Temp. Gyrus	42.8	−52.4	20.7	353
Frontal	L Precentral Gyrus	−27.2	−11.0	56.4	1987
	L Mid. Front. Gyrus	−27.0	34.0	39.0	739
	L Med. Front. Gyrus	−14.6	46.5	10.4	270
	*R Mid. Front. Gyrus	23.9	−6.5	55.4	4938
	R Sup. Front. Gyrus	25.3	27.4	48.9	1149
Parietal	R Inf. Front. Gyrus	47.5	3.8	32.2	800
	R Precentral Gyrus	38.4	24.8	37.4	548
	L Postcentral Gyrus	−52.4	−28.3	33.8	279
Occipital	L Lingual Gyrus	−9.3	−72.6	−2.9	383
	*R Precuneus	15.1	−64.4	35.4	73,662
	R Inf. Occip. Gyrus	31.4	−83.4	−10.7	544
Cingulate	R Anterior Cingulate	3.0	40.2	6.2	698

^a Talairach coordinates of cluster center of mass.

in left temporal and frontal regions, including the left IFG and left MTG, along with smaller clusters in the right hemisphere (Fig. 2; Table 4).

Finally, we examined activation associated with semantic processing by examining the conjunction of WORD MEANING > WORD CASE and WORD MEANING > WORD RHYME contrasts (Fig. 3; Table 5). The largest clusters of meaning-related activation were located in bilateral middle frontal and inferior parietal regions. An additional cluster was observed in the left middle temporal region.

3.2. Relationships between individual differences in reading and reading-related activation

The analyses presented above identified distinct patterns of activation associated with different types of reading subprocesses. We next explored how individual differences in activation associated with reading words relate to individual differences in reading

ability. For each offline measure of reading ability, we calculated correlations with the degree of activation during all conditions in which participants viewed words, whether making case, rhyme, or meaning judgments. Thus, individual differences in reading ability were associated with activation during word reading irrespective of task type. In the following analyses, we investigated regions in which associations exist between the degree of activation during the processing of familiar words and performance on standardized measures.

We first examined how individual differences in orthographic processing, as indexed by the Sight Word Efficiency subtest of the TOWRE, were related to activation during word reading (Table 6). The largest cluster of association was located on the medial surface at the junction of the occipital and parietal lobes (Fig. 4A). In this region, greater familiar word reading ability was associated with greater activation during word reading. A similar association was present in other portions of the medial occipital/parietal junction. In contrast, a negative association was seen in a portion of the left MTG (Fig. 4B). In this region, greater differences in activation between word reading tasks and fixation were associated with worse performance on the standardized measure of familiar word identification.

We next examined relationships between individual differences in phonological decoding skill, as indexed by the Phonemic Decoding Efficiency subtest of the TOWRE, and the patterns of activation during word reading. Greater activation during word reading was associated with greater phonological decoding ability in some cortical regions (Table 7). These regions included the left supramarginal gyrus (Fig. 5A) and left MTG (Fig. 5B). Individuals with greater differences in activation between word reading tasks and fixation in these regions showed better performance on the Phonemic Decoding Efficiency subtest of the TOWRE.

To explore relationships between individual differences in reading comprehension ability and degree of word-related activation, we examined associations between scaled scores on the Nelson-Denny Reading Comprehension test and beta-weights from the contrast of word tasks with fixation (Table 8). In the left anterior cingulate (Fig. 6A) and left MTG (Fig. 6B), greater word-related

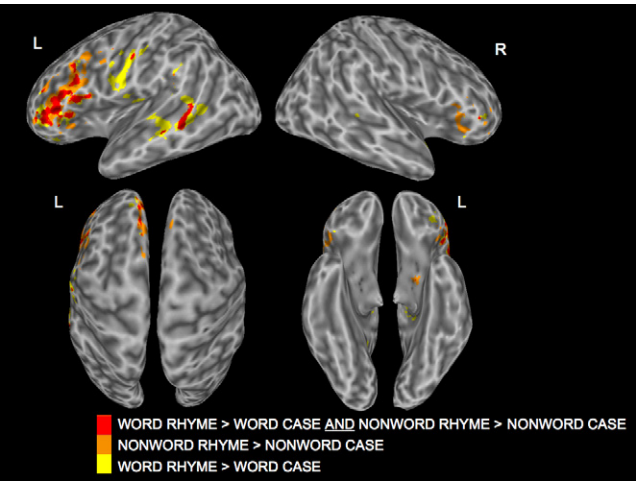


Fig. 2. Regions showing greater activation in WORD RHYME > WORD CASE and PSEUDOWORD RHYME > PSEUDOWORD CASE contrasts ($p < .005$) displayed on a cortical surface model of the N27 atlas brain.

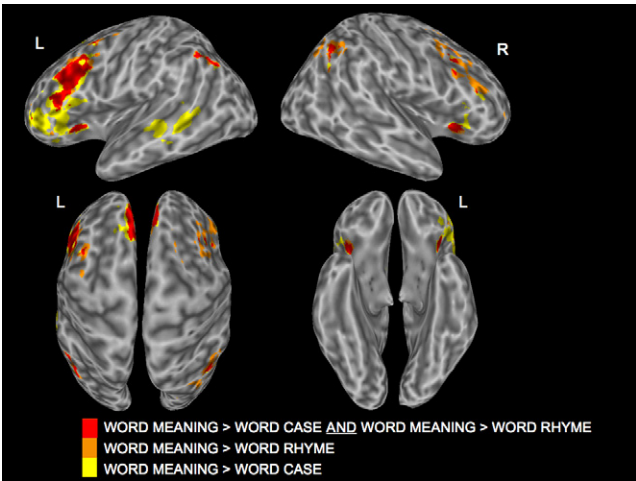


Fig. 3. Regions showing greater activation in WORD MEANING > WORD RHYME and WORD MEANING > WORD CASE contrasts ($p < .005$) displayed on a cortical surface model of the N27 atlas brain.

activation was associated with worse performance on the Nelson-Denny Reading Comprehension test.

4. Discussion

The first goal of the present study was to identify brain regions associated with orthographic, phonological, and semantic components of reading in typical adults. We independently manipulated the characteristics of stimuli (e.g., words versus consonant strings in case decisions) and the task demands (e.g., rhyme versus case decisions on the same stimuli) in order to reduce potential confounds in either task demands or stimulus characteristics. As discussed below, we identified patterns of brain activity engaged by phonological, orthographic, and semantic processing. These patterns of activation are broadly consistent with prior studies identifying regions associated with specific component skills of reading. More importantly to the current study, they demonstrate that manipulating task demands and stimulus characteristics independently can modulate activity in component brain regions.

We first investigated brain regions associated with orthographic processing. The first analysis identified regions sensitive to actively processing letter case versus rhyme judgments on the same words. Of note, the same stimuli were used in both sets of trials such that observed activity cannot be attributed to the pronounceability or other phonological characteristics of

Table 5
Regions demonstrating greater activation during WORD MEANING > WORD CASE and WORD MEANING > WORD RHYME contrasts.

Anatomical location		Talairach coordinates ^a			Size (mm ³)
		x	y	z	
Temporal	L Mid. Temp. Gyrus	−53.8	−35.5	1.0	2612
	L Mid. Front. Gyrus	−42.3	19.7	26.7	25,699
Frontal	L Sup. Front. Gyrus	−3.1	22.5	51.1	10,672
	R Precentral	37.6	16.0	34.1	10,777
Parietal	R Sup. Front. Gyrus	32.8	56.6	16.0	865
	L Inf. Par. Lob.	−44.7	−63.5	42.3	2581
	R Inf. Par. Lob.	38.1	−62.5	39.4	3793

^a Talairach coordinates of cluster center of mass.

the stimuli. Case judgments resulted in greater activation in bilateral occipital cortex and portions of frontal cortex.

The second analysis was interested in a somewhat different aspect of orthographic processing related to knowledge of how letters are combined to form legal words in English (Olson, Kliegl, Davidson, & Foltz, 1985). We contrasted activation during case judgment of familiar words and orthographically legal pseudowords to case judgment of orthographically illegal consonant strings. No region showed greater activation to all orthographically legal stimuli (words and pseudowords); however, left frontal re-

Table 4
Regions demonstrating greater activation during WORD RHYME > WORD CASE and PSEUDOWORD RHYME > PSEUDOWORD CASE contrasts.

Anatomical location		Talairach coordinates ^a			Size (mm ³)
		x	y	z	
Temporal	L Mid. Temp. Gyrus	−54.2	−36.0	2.3	3343
	R Sup. Temp. Gyrus	43.5	10.9	−9.6	812
Parahippocampal	L Parahippocampal	−20.2	−3.8	−11.9	341
	R Parahippocampal	26.4	−44.7	10.3	3257
Frontal	L Inf. Front. Gyrus	−44.9	23.9	17.3	17,614
	L Sup. Front. Gyrus	−4.8	26.7	54.2	5297
	L Precentral	−53.5	−9.8	32.0	2551
	L Sup. Front. Gyrus	−6.6	−4.7	68.4	315
	R Inf. Front. Gyrus	43.8	22.6	7.4	1326
	R Inf. Front. Gyrus	44.8	40.2	5.8	612
Parietal	L Inf. Par. Lob.	−56.1	−39.0	23.5	339
Insula	L Insula	−37.8	−20.0	22.9	592
Cingulate	R Cingulate	13.1	−14.6	40.5	336

^a Talairach coordinates of cluster center of mass.

Table 6

Regions demonstrating associations between activation and SWE scaled score.

Anatomical location		Talairach coordinates ^a			Size (mm ³)
		x	y	z	
<i>Positive correlations</i>					
Occipital	L Precuneus	−7.1	−52.9	37.2	336
	R Precuneus	3.2	−68.9	28.9	1827
	R Cuneus	6.4	−75.7	6.9	1666
<i>Negative correlations</i>					
Temporal	L Mid. Temp. Gyrus	−50.2	−52.8	−6.8	365

^a Talairach coordinates of cluster center of mass.**Table 7**

Regions demonstrating associations between activation and PDE scaled score.

Anatomical location		Talairach coordinates ^a			Size (mm ³)
		x	y	z	
<i>Positive correlations</i>					
Temporal	L Mid. Temp. Gyrus	−55.5	−62.9	12.2	343
Frontal	L Med. Front. Gyrus	−11.7	45.4	18.5	401
Parietal	L Supramarginal	−59.0	−45.7	34.7	338
Cingulate	L Ant. Cingulate	−9.8	29.5	17.4	337
	L Ant. Cingulate	−10.9	15.1	−6.6	284
	R Ant. Cingulate	10.6	32.4	22.0	353

^a Talairach coordinates of cluster center of mass.

gions showed greater activation for familiar words than consonant strings and left temporal regions showed greater activation for orthographically legal pseudowords than consonant strings. While

it is possible to conclude that these regions represent areas of specialization for familiar word forms, we also acknowledge that, as with other studies focusing on words versus illegal consonant

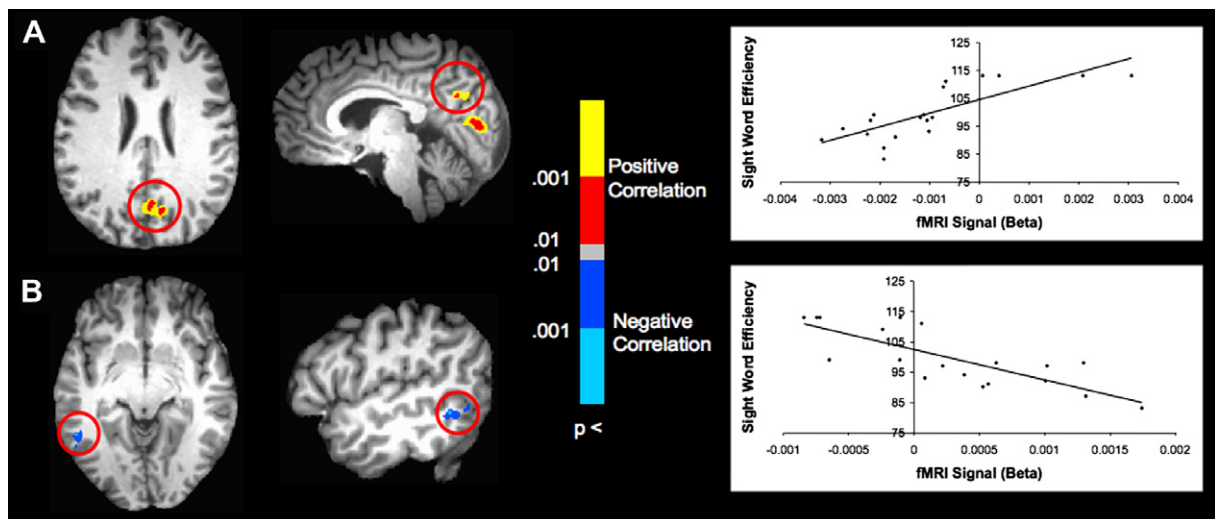


Fig. 4. Correlation between activation elicited by words (orthographic, phonological and semantic judgments) and scaled score on the TOWRE SWE subtest. Positive relationships (greater activation with higher scores) are shown in red/yellow, and negative relationships (greater activation with lower scores) are shown in blue. Scatterplots illustrate relationships between beta values from the contrast and scaled scores on the TOWRE SWE subtest in (A) Right medial occipital cortex and (B) Left middle temporal gyrus.

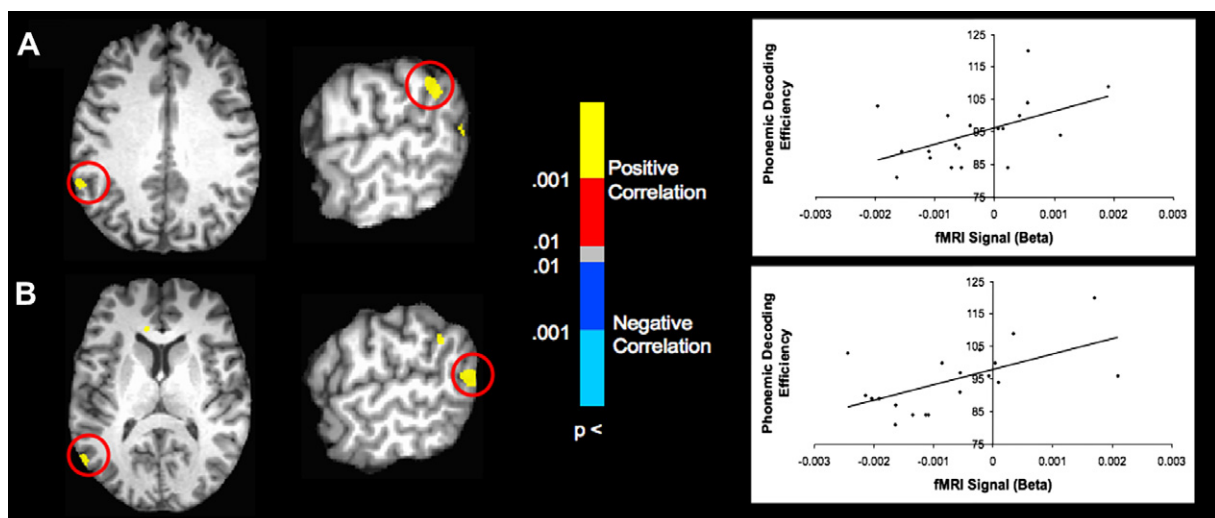


Fig. 5. Correlation between activation elicited by words (orthographic, phonological and semantic judgments) and scaled score on the TOWRE PDE subtest. Positive relationships (greater activation with higher scores) are shown in red/yellow, and negative relationships (greater activation with lower scores) are shown in blue. Scatterplots illustrate relationships between beta values from the contrast and scaled scores on the TOWRE PDE subtest in (A) Left parietal lobe (B) Left insula.

Table 8

Regions demonstrating associations between word-reading activation and Nelson-Denny Comprehension scaled score.

Anatomical location		Talairach coordinates ^a			Size (mm ³)
		x	y	z	
<i>Positive correlations</i>					
Temporal	L Fusiform	−18.0	−87.9	−13.4	564
Cingulate	R Cingulate	23.7	−18.3	27.7	556
<i>Negative correlations</i>					
Temporal	L Mid. Temp. Gyrus	−52.8	−63.3	1.1	276
Frontal	L Precentral Gyrus	−46.3	−13.2	53.3	260
Parietal	L Postcentral Gyrus	−49.0	−18.8	15.8	320
Cingulate	L Ant. Cingulate	−0.7	25.2	−2.3	1475

^a Talairach coordinates of cluster center of mass.

strings, (Buchel et al., 1998; Cohen et al., 2002; Price et al., 1996), these differences could instead reflect pronouncability, or perhaps semantic content of real words in the case of the WORD CASE > CONSONANT STRING CASE contrast.

To identify brain activity patterns associated with phonological processing, we contrasted activation during rhyme judgments of words and pseudowords to activation during case judgments of identical stimuli. Portions of the left MTG and right STG showed preferential activation to both types of rhyme judgments, consistent with results from prior studies (Lurito, Kareken, Lowe, Chen, & Mathews, 2000; Pugh et al., 2000; Shaywitz et al., 1995). The left inferior prefrontal cortex also showed greater activation for phonological decisions. In our study, IFG activation extended into both the anterior portion (BA 45/47/10) that has been more often associated with semantic processing and the posterior portion (BA 44/6), that has been more often associated with phonological processing (Bokde, Tagamets, Friedman, & Horwitz, 2001; Fiez, 1997; Poldrack et al., 1999). This result supports the TMS finding that anterior and posterior portions of the left prefrontal cortex may both contribute to phonological processing, but do so to different extents (Devlin et al., 2003).

Finally, we identified areas of specialization for semantic processing by contrasting activation during meaning judgments of words to activation during case and rhyme judgments of the identical stimuli. Results yielded greater activity in a wide range of cortical regions including middle temporal, inferior parietal and

frontal gyri, as predicted by prior studies of semantic judgments of word stimuli (McDermott et al., 2003; Pugh et al., 1996). It should be acknowledged, however, that a broader set of semantic judgments might have revealed different patterns of activation. Much work has characterized finer-grained questions of the organization of semantic knowledge, including neural correlates of semantic priming (Rissman, Eliassen, & Blumstein, 2003; Rossell, Price, & Nobre, 2003), processing of abstract versus concrete concepts (Binder, Westbury, McKiernan, Possing, & Medler, 2005), and processing of pictures versus words (Chao, Haxby, & Martin, 1999; Perani et al., 1999). Here we sought only to identify regions broadly engaged in accessing word meaning, and set aside questions related to whether this network can be further subdivided along parameters such as animacy or concreteness.

4.1. Brain-behavior relationships

Our study also had a second complementary aim, which was to investigate relationships between reading-related activation and different aspects of reading skill. This approach sought to provide additional insight into the association between patterns of reading-related neural activity and orthographic, phonological, and semantic processes. Relationships with reading-related activation varied across standardized measures tapping specific reading skills, suggesting that different patterns of activation are associated with different aspects of reading. Familiar word identification, phonological decoding, and reading comprehension ability were each associated with activation during the word reading tasks, but the regions of association varied with reading subskill. Negative correlations between word-related activation and speeded identification of familiar words were observed in a portion of the left MTG, while positive associations were seen in occipital regions. Positive associations between word-related activation and speeded phonological decoding of pseudowords were seen in the left temporo-parietal regions. Negative relationships between activation and reading comprehension ability were observed in the widespread bilateral temporal, parietal, and frontal regions.

Next, we discuss relationships between each reading skill and reading-related activation in more detail. With respect to orthographic processing, the degree of activation during word processing was positively associated with activation in occipital cortex.

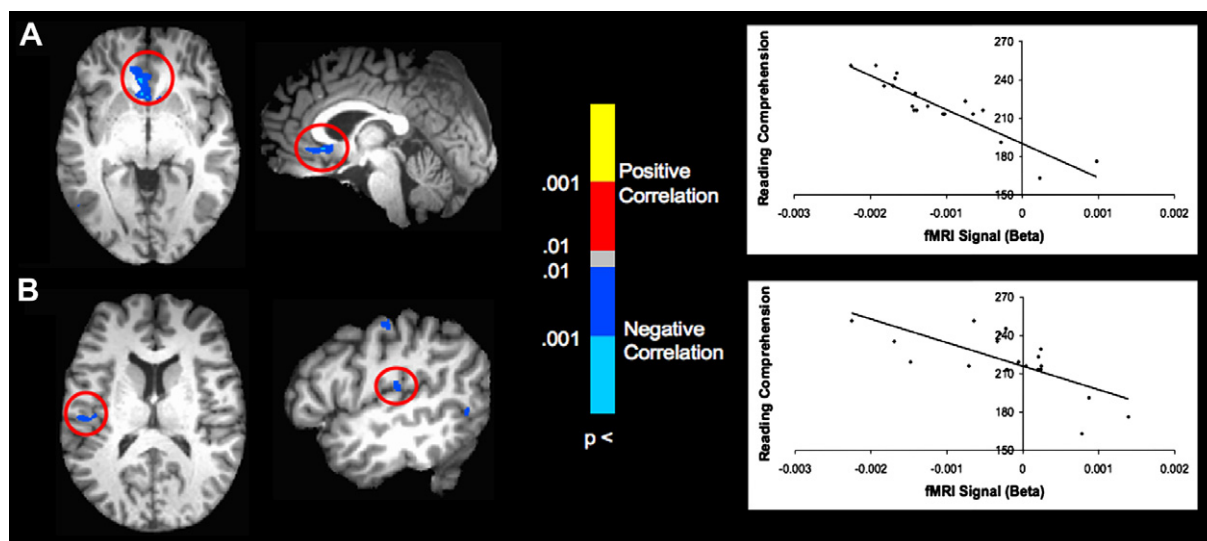


Fig. 6. Correlation between activation elicited by words (orthographic, phonological and semantic judgments) and scaled score on the Nelson-Denny Reading Comprehension subtest. Positive relationships (greater activation with higher scores) are shown in red/yellow, and negative relationships (greater activation with lower scores) are shown in blue. Scatterplots illustrate relationships between beta values from the contrast and scaled scores on the TOWRE PDE subtest in (A) anterior cingulate (B) Left temporal cortex.

Conversely, the amount of activation while reading words was negatively correlated with sight word reading ability in a portion of the left MTG. Those participants whose MTG activation was greater performed less well on the SWE subtest, a speeded measure of familiar word recognition. One possibility is that enhanced engagement of this region reflects more effortful, less automatized word recognition, such that individuals who show the strongest activation in this region are those in whom sight word recognition is least efficient. We observed no significant relationship between activation in fusiform cortex and efficiency of sight word recognition. Prior studies have been inconsistent with regard to such relationships, with one finding that letter recognition accuracy was negatively related to activation level in this region in typical adult readers (Garrett et al., 2000) and others finding the opposite effect such that fusiform cortex is under-activated in either dyslexic readers vs. controls (Brunswick et al., 1999). Thus, we cannot rule out the possibility that engagement of this region is associated with better performance of some reading-related tasks.

Associations with phonological processing ability were observed in different brain regions. Relationships between speeded nonword reading and activation were observed in the left inferior parietal lobule and MTG. This finding is consistent with the observation that in typical adult readers, better on-line rhyme performance was associated with greater activation in superior temporal and supramarginal/angular gyri (Booth et al., 2002). These associations are also in harmony with prior studies demonstrating reduced activation of left temporo-parietal areas during phonological processing in children with (or at risk for) dyslexia (Papanicolaou et al., 2003; Simos et al., 2000b; Temple et al., 2001) and in compensated dyslexic adults (Paulesu et al., 1996).

Given prior studies identifying atypical activation in IFG in individuals with dyslexia (Georgiewa et al., 1999; Papanicolaou et al., 2003; Temple et al., 2001), it is perhaps surprising that we did not observe associations between variation in phonological decoding ability and activation in the IFG. Across participants we observed large clusters of activation in left IFG during the phonological judgment task. However, activation in this region was not correlated with individuals' phonological decoding performance. In spite of the significant range of nonword reading ability in our sample (Table 1), and demonstrated relationships in temporo-parietal cortex, we failed to find associations between variability in this skill and activation in inferior frontal cortex. This result appears to support the theory that dysfunction in this region is specifically associated with frank reading impairment that occurs outside of the range of typical reading ability. That is, reduced activity in IFG during phonological tasks in dyslexia may reflect a marked discontinuity with decoding skill in typical readers. We also note some recent work indicating altered brain anatomy (reduced radial expansion, a measure of local brain size) in frontal cortex in adults with both poor comprehension and decoding, but not in those with poor phonological decoding but normal comprehension (Welcome, Chiarello, Thompson, & Sowell, 2011), supporting the idea that alterations in frontal regions are associated with truly impaired reading, rather than with individual differences within the normal reading population.

Associations with semantic processing ability were observed in portions of temporal, frontal, and parietal cortex as well as in left anterior cingulate. These regions generally showed an inverse relationship between degree of reading-related activation and reading comprehension skill. Those individuals with the greatest reading-related activation in these regions performed worse on a measure of reading comprehension. It is interesting to note that *positive* associations with phonological decoding ability were observed in nearby brain regions in our study. The fact that we have found relationships in opposite directions for the two subskills supports the view of a trading relationship between decompositional

(phonological) and holistic (lexical/semantic) knowledge of words in normal readers (Harm & Seidenberg, 2004). That is, it supports the view that readers differ in the degree to which they employ analytic versus holistic strategies during reading, and in such a way that the relative engagement of these pathways varies as a function of reading ability.

4.2. Task contrasts vs. individual differences

A key finding of the present study is that our approaches to subdividing the regions engaged by reading yielded only partially overlapping results. There was some overlap between the orthographic, phonological, and semantic regions identified in the groupwise contrasts and the regions in which individual reading skills were correlated with reading-related activation. For example, the left MTG was differentially active during phonological processing and activation varied with phonological decoding skill. Similarly, some of the regions showing greater activation during the processing of word meanings were regions in which activation varied with reading comprehension. Where this overlap exists, such that a specific type of processing engages the region and skill in that domain modulates the level of activation, it is relatively straightforward to link functional activation with reading sub-processes.

In contrast, other regions were identified as specifically sensitive to orthographic, phonological, or semantic processing in the first analyses, but did not vary with respect to individual differences in the corresponding reading skill. For instance, inferior frontal cortex was identified in the phonological contrast, but degree of word-related activation within this region did not vary with phonemic decoding skill. The explanation appears to be that activation within some regions reflects processes that are essential for skilled reading, and the failure to fully engage these cortical regions thus represents a frank deviation from 'typical reading'. Since the individuals in this study were typical reading adults, it is possible that we did not observe relationships between reading ability and activation in some regions because most or all participants showed typical activation in these regions.

4.3. Individual differences vs. development and dyslexia

Our results also address the question of whether individual differences in typical reading adults mirror changes in the reading network associated with development or with reading impairment. Under one view, adults with less reading skill might show patterns of activation similar to those of younger children, reflecting a type of protracted developmental delay. Likewise, if dyslexia simply represents one end of the spectrum of reading skill, it would be predicted that differences in functional brain activity associated with dyslexia would be reflected in variability observed within typical readers as well. Admittedly, some findings from the present study are consistent with these possibilities such as the finding that temporo-parietal activation during phonological processing varied with phonemic decoding ability. However the overall pattern of results suggests an alternative hypothesis, that normal variability in reading skills is associated with qualitatively different patterns of activation than that which is observed in either dyslexia or typical reading development. While it is possible that our results do not replicate prior studies using developmental or impaired populations because of reduced power or other methodological issues, it is worth noting that our sample showed a great deal of variability in all three standardized reading measures (Table 1), and power was sufficient to observe associations in some regions. Therefore, it is intriguing to speculate that individual differences in reading skill in adults relate to a partially different network of cortical regions than those that show changes with reading

development or atypical activation in dyslexics. Further, different relationships with activation were associated with different reading subskills (efficiency of familiar word identification, Phonemic Decoding Efficiency, and reading comprehension), suggesting that different aspects of reading skill are associated with activation in different portions of cortex. Understanding such relationships between reading activation and behavior in typical-reading adults is necessary in order to obtain a full picture of the cortical networks subserving skilled reading.

5. Conclusions

The present study identified patterns of activation associated with subcomponents of visual word reading in a sample of typical adult readers. Next, we explored how performance on behavioral measures of component reading skills was related to activation in different cortical regions. Phonological processing engaged left inferior frontal and temporal regions. Semantic processing was associated with increased activation in bilateral middle frontal and parietal regions. Individual differences in familiar word identification were associated with individual differences in activation in occipital cortex and left MTG. Phonological decoding skill was not associated with variation in inferior frontal cortex, but with activation in temporo-parietal regions. Finally, reading comprehension skill was associated with decreased activation in the anterior cingulate. These results suggest that while a core group of regions may be engaged by skilled readers while processing text, there is meaningful variation in the cortical regions used by different individuals. Importantly, degree of activation associated with reading related differently to different aspects of reading skill, and regions that correlated with cognitive measures of reading subskills failed to map in a one-to-one fashion with the subregions that were identified in the initial analysis. Moreover, we identified only partial overlap between brain regions associated with normal variability in phonological decoding skills compared to those previously identified as associated with typical reading development or with dyslexia. This finding indicates that individual differences in familiar word identification and reading comprehension have partially unique relationships with reading-related brain activation, and that variability in brain/behavior relationships in adults do not simply reflect the delayed maturation of less-skilled readers or simply the lower end of a continuum of reading skill.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.bandl.2011.12.011](https://doi.org/10.1016/j.bandl.2011.12.011).

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